

Tennessee Comprehensive Assessment Program

TCAP

Science
Grade 5
Practice Items



The distances of several stars from Earth are shown in the table.

Star Distances

Star	Distance from Earth (light-years)
Capella	43
Sirius	9
Polaris	434
Spica	261

Which of these describes what will be seen when looking at the stars from Earth?

- A. Capella appears brighter than Sirius but dimmer than Polaris when viewed from Earth.
- B. Spica appears dimmer than Polaris but brighter than Capella when viewed from Earth.
- C. Capella appears brighter than Spica but dimmer than Sirius when viewed from Earth.
- D. Polaris appears brighter than Spica but dimmer than Capella when viewed from Earth.

A toy car moves at a constant rate along a level surface. A student measures the distance the toy car moves in centimeters every ten seconds. The measurements are recorded in the data table.

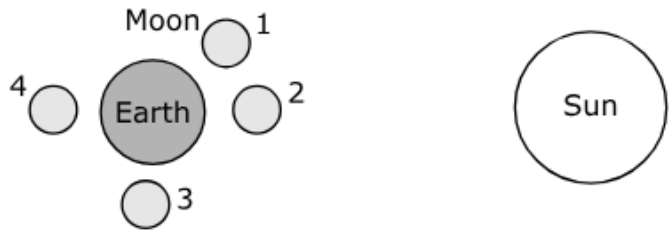
Toy Car Motion

Measurement	1	2	3	4	5	6
Distance (cm)	4	8	12	16	___?	___?
Time (s)	10	20	30	40	50	60

Which **two** distances are the **best** predictions to complete the table?

- M. 18 cm
- P. 20 cm
- R. 24 cm
- S. 28 cm
- T. 30 cm

The diagram shows Earth, the sun, and four positions the moon might be in as it revolves around Earth.

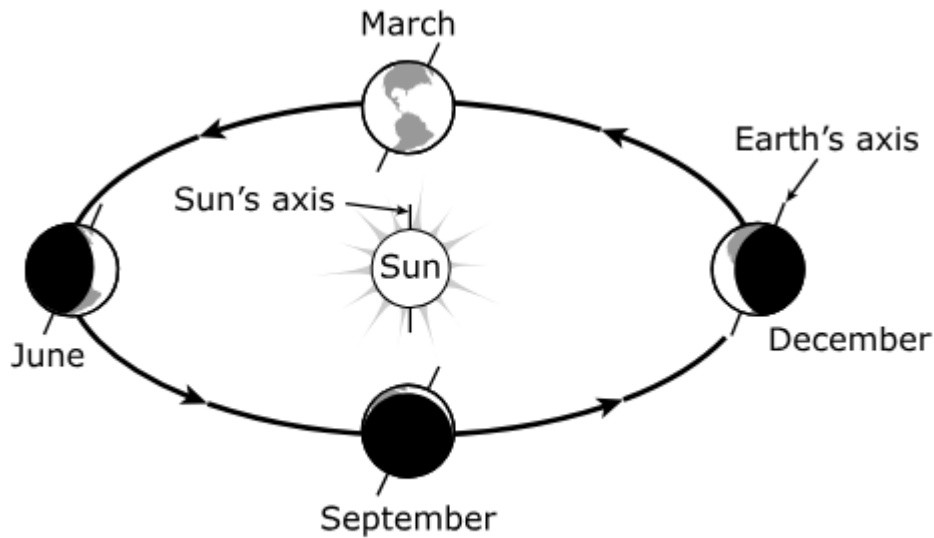


Note: Figure is not drawn to scale.

A solar eclipse is **most likely** to occur when the moon is in which position?

- A. position 1
- B. position 2
- C. position 3
- D. position 4

Parts of Earth receive different amounts of daylight during the year. The diagram shows Earth in different positions in its revolution around the sun.

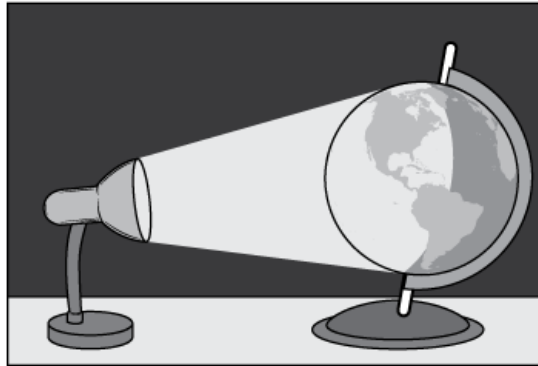


Note: Figure is not drawn to scale.

The changes in the number of daylight hours on Earth are caused by

- A. the tilt of Earth's axis.
- B. the tilt of the sun's axis.
- C. the distance of Earth from the sun.
- D. the movement of the sun around Earth

Students use a desk lamp to model light from the sun. The students use a globe to model Earth. The students shine the light from the desk lamp onto the globe.



The students want to model the reason that the sun appears to rise and set on Earth.

Which of these should the students do with the parts of the model?

- A. Spin the globe on its stand.
- B. Turn the desk lamp on and off.
- C. Lift the desk lamp above the globe.
- D. Carry the globe in a circle around the desk lamp.

Students place four bean seeds into a pot. All of the seeds sprout roots and shoots. The plants have green leaves and stems with white flowers. The mature plants produce long bean pods.

Which **three** plant characteristics are **most likely** a result of interactions with the environment?

- A. All of the seeds sprout roots that grow down.
- B. If the plants are not watered, they start to droop.
- C. The mature plants produce long green pods containing beans.
- D. The plants have green stems, green leaves, and white flowers.
- E. The shoots all bend to grow toward the only source of sunlight.

Students wanted to learn what happens when substances are mixed. They mixed liquids and solids for their experiment. Their observations are recorded in the table.

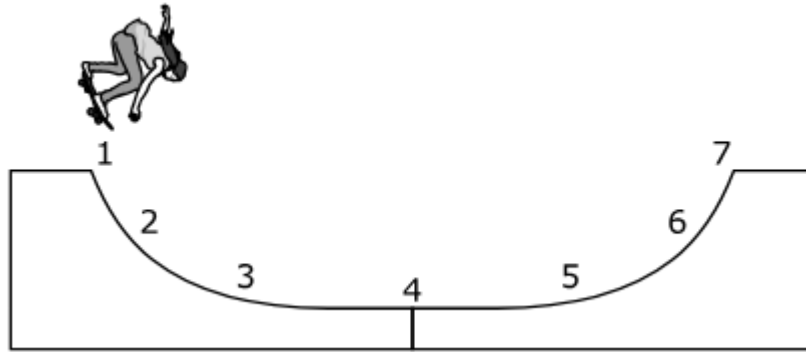
Students’ Data

Mixture	Liquid	Solid	Observations
1	Water (no color)	Sugar (white)	Forms a liquid with no color
2	Vinegar (no color)	Baking soda (white)	Gas bubbles out of a liquid with no color

Which mixture went through changes that produced new substances?

- M. Mixture 1, because it formed only a liquid
- P. Mixture 1, because it had no color
- R. Mixture 2, because bubbles formed
- S. Mixture 2, because no solid formed

A local skate park has a U-shaped ramp, as shown. A student rides her skateboard back and forth along the ramp at a steady pace. She reaches the next location every 0.4 seconds.



The student starts a timer when she leaves Location 1. How many seconds will it take her to return to Location 1?

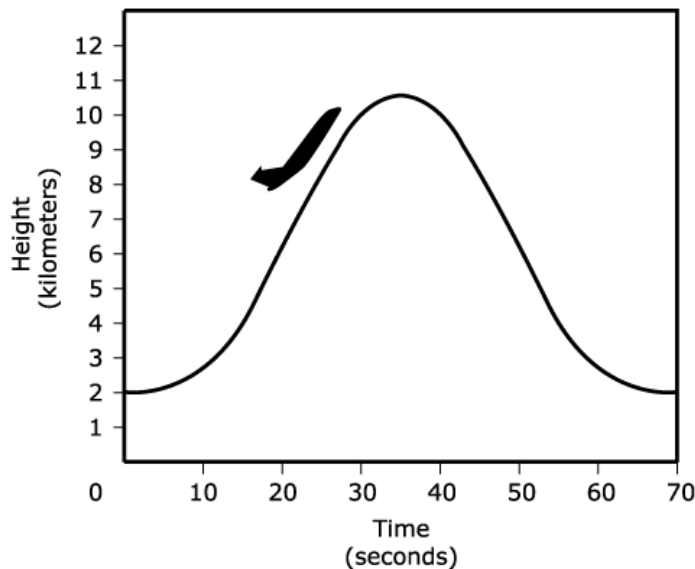
- M. 4.4 seconds
- P. 4.6 seconds
- R. 4.8 seconds
- S. 5.6 seconds

The questions on pages 11 -12 refer to the passage(s) and image(s) shown.

Astronaut Training – Part 1

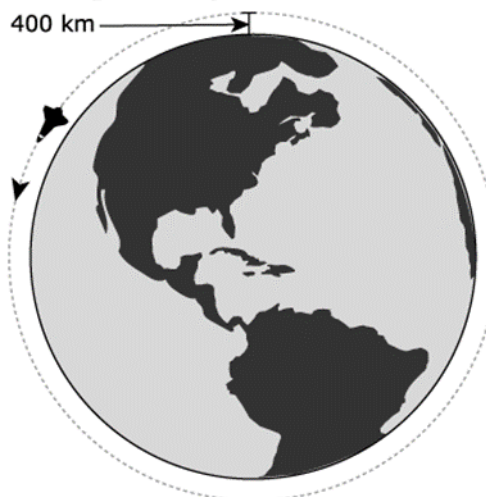
One way that astronauts train for spaceflight is on an airplane that flies up and down in steep curves. Astronauts train for about an hour each day on the airplane while it flies up and down as many as 60 times. Figure 1 shows how the airplane flies. Astronauts start to feel weightless at the point in the flight shown by the picture of the airplane.

Figure 1. Astronaut Training Airplane



Actual spaceflights usually take astronauts into orbit, where they may stay for a few days or even several months. Astronauts in orbit circle Earth as many as 16 times each day. Astronauts feel weightless any time they are in orbit. Figure 2 shows an example of a spacecraft in orbit around Earth.

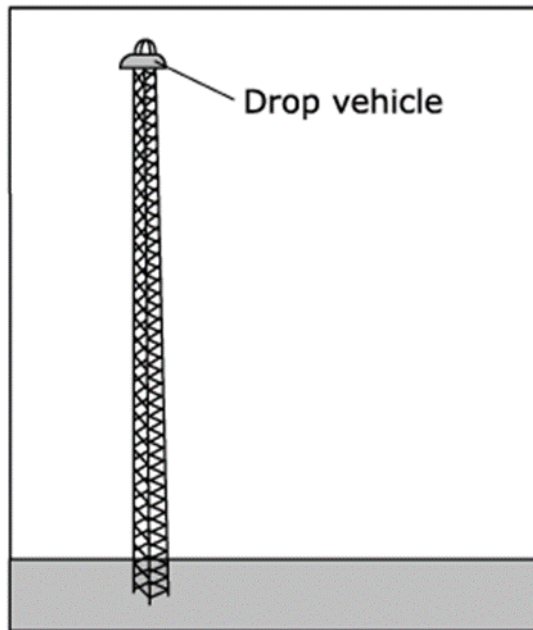
Figure 2. Spacecraft in Orbit



Astronaut Training – Part 2

People who want to feel weightless like an astronaut can ride on a drop tower. The ride takes people to the top of a tall tower. The drop vehicle and the people fall for about three seconds. Near the bottom of the tower, the drop vehicle slows to a stop. Figure 3 shows the drop tower with the drop vehicle at the top of the tower.

Figure 3. Drop Tower



1. Which statement **best** compares the force of gravity acting on the airplane in Figure 1 (Part 1) and the spacecraft in Figure 2 (Part 1)?
 - M. the force of gravity on the plane is upwards while the force of gravity on the spacecraft is forwards
 - P. the force of gravity is towards the center of the earth for both the airplane and the spacecraft
 - R. the force of gravity is forwards for the airplane and upwards for the spacecraft
 - S. the force of gravity is towards the surface of the earth for both the airplane and the spacecraft

2. Which claim is **most** supported by evidence in Figure 1 and Figure 2 (Part 1)?
 - A. Earth's gravity only affects objects on the surface of Earth.
 - B. An object being pulled toward Earth's center can move in a curved path.
 - C. Earth only pulls on objects when they are moving away from the surface of Earth.
 - D. Earth's gravity is stronger in space.

3. Based on Figure 3 (Part 2), which statement **best** describes why the people riding the drop tower can feel weightless?
 - M. The people and the drop vehicle are both moving in the direction that gravity pulls.
 - P. The people and the drop vehicle are both moving too quickly for gravity to affect them.
 - R. The force of gravity decreases because the people and the drop vehicle move in opposite directions.
 - S. The force of gravity between Earth and both the people and the drop vehicle is too small to affect them.

4. When the drop vehicle approaches the bottom of the drop tower, the drop vehicle begins slowing down. During this time, the force on the people is much greater than the force of gravity. Based on Figure 3 (Part 2), which statement **best** describes a reason for this increase in force?
- A. The drop vehicle is at the closest point in its drop to Earth's surface.
 - B. The force slowing the drop vehicle is in a direction that is away from Earth's center.
 - C. The force of gravity on the people increases because the drop vehicle is moving toward Earth's surface.
 - D. The force of gravity starts to pull on the people when the drop vehicle begins moving away from Earth's center.

Metadata - Science Items

Page Number	Item Type	Key	TN Standards
1	MC	C	5.ESS1.1
2	MS	P, R	5.PS2.2
3	MC	B	5.ESS1.4
4	MC	A	5.ESS1.5
5	MC	A	5.ESS1.3
6	MS	A, B, E	5.LS3.1
7	MC	R	5.PS1.3
8	MC	R	5.PS2.2

Astronaut Training

Page Number	Item Type	Item Number	Key	TN Standards
9	Stimulus			
10	Stimulus			
11	MC	1	P	5.PS2.3
11	MC	2	B	5.PS2.3
11	MC	3	M	5.PS2.3
12	MC	4	B	5.PS2.1

Metadata Definitions

Item Type	MC = Multiple Choice MS = Multiple Select Stimulus = Cluster support information
Key	Correct Answer
TN Standards	Primary standard assessed in item.